ERD Design, Normalization, Data Integrity Rules

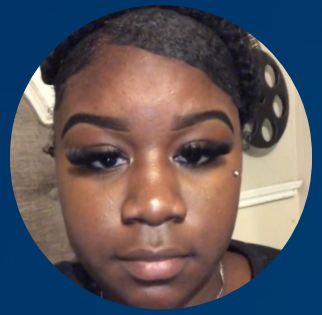
SAFE BANK - CREDIT CARD MANAGEMENT SYSTEM

START

TEAM



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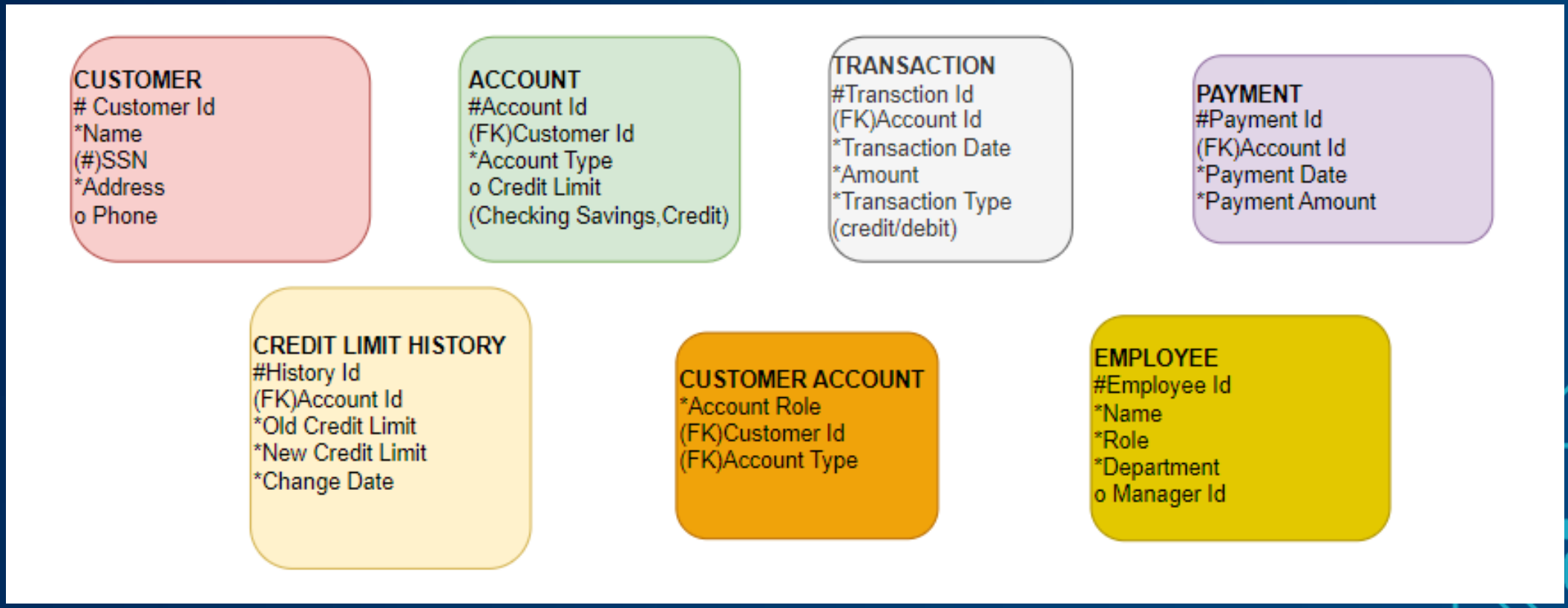
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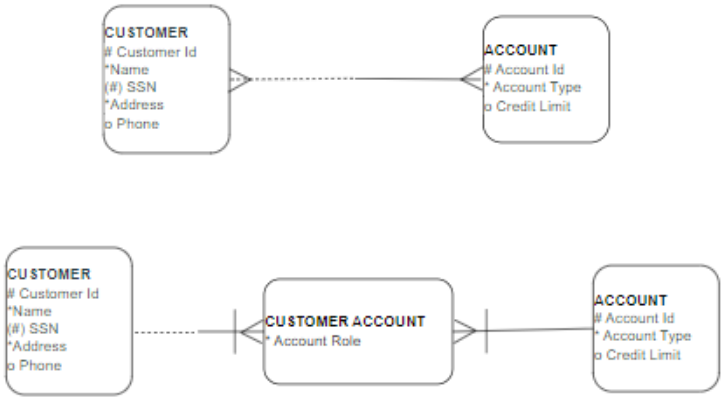
- Goal
- Team
- Entities & Attributes
- Intersection Entity
- Recursive Relationship
- Credit Limit History
- ERD (Part 1)
- Relationship Matrix
- ERD Final
- Normalization
- Data Integrity Rules
- Thanks

GOAL

THE PROJECT FOCUSES ON CREATING A DATABASE SYSTEM FOR SAFEBANK TO EFFICIENTLY MANAGE BANKING OPERATIONS AND ENSURE DATA ACCURACY. IT AIMS TO SUPPORT THE BANK'S NEEDS FOR ORGANIZATION AND RELIABLE RECORD-KEEPING.

ENTITIES AND ATTRIBUTES



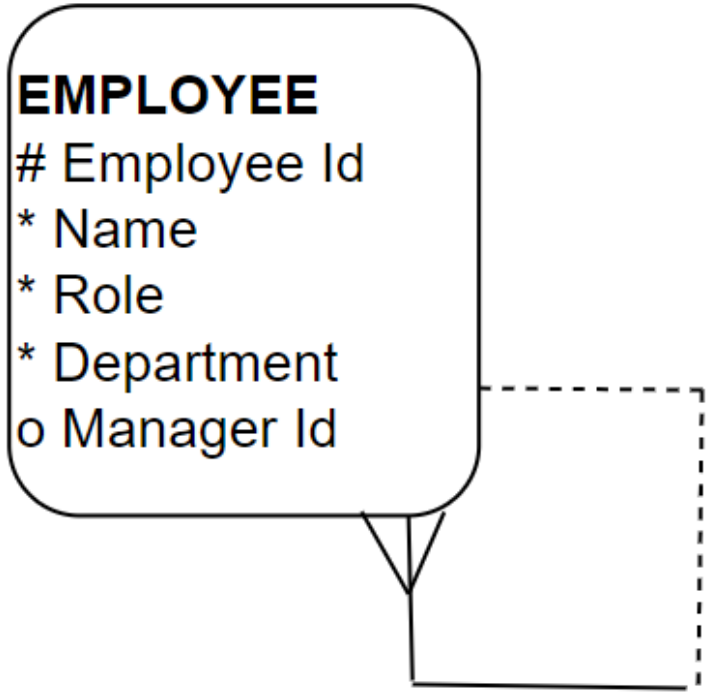


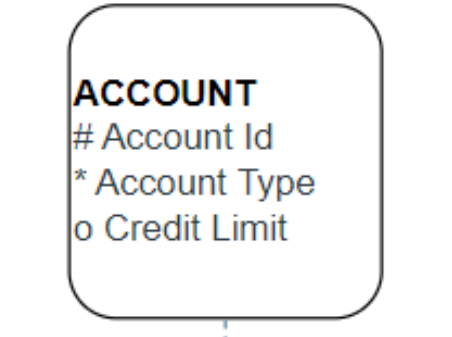
INTERSECTION ENTITY

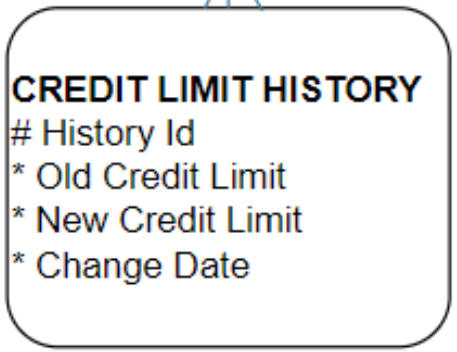
Customer and Account has a many to many relationship. To solve this, we create the intersection table Customer Account using a barred relationship. It will manage joint ownership to bridge together the tables.

RECURSIVE RELATIONSHIP

Each Employee may manange one or more Employees, but Each Employee must be managed by one and only one Employee.

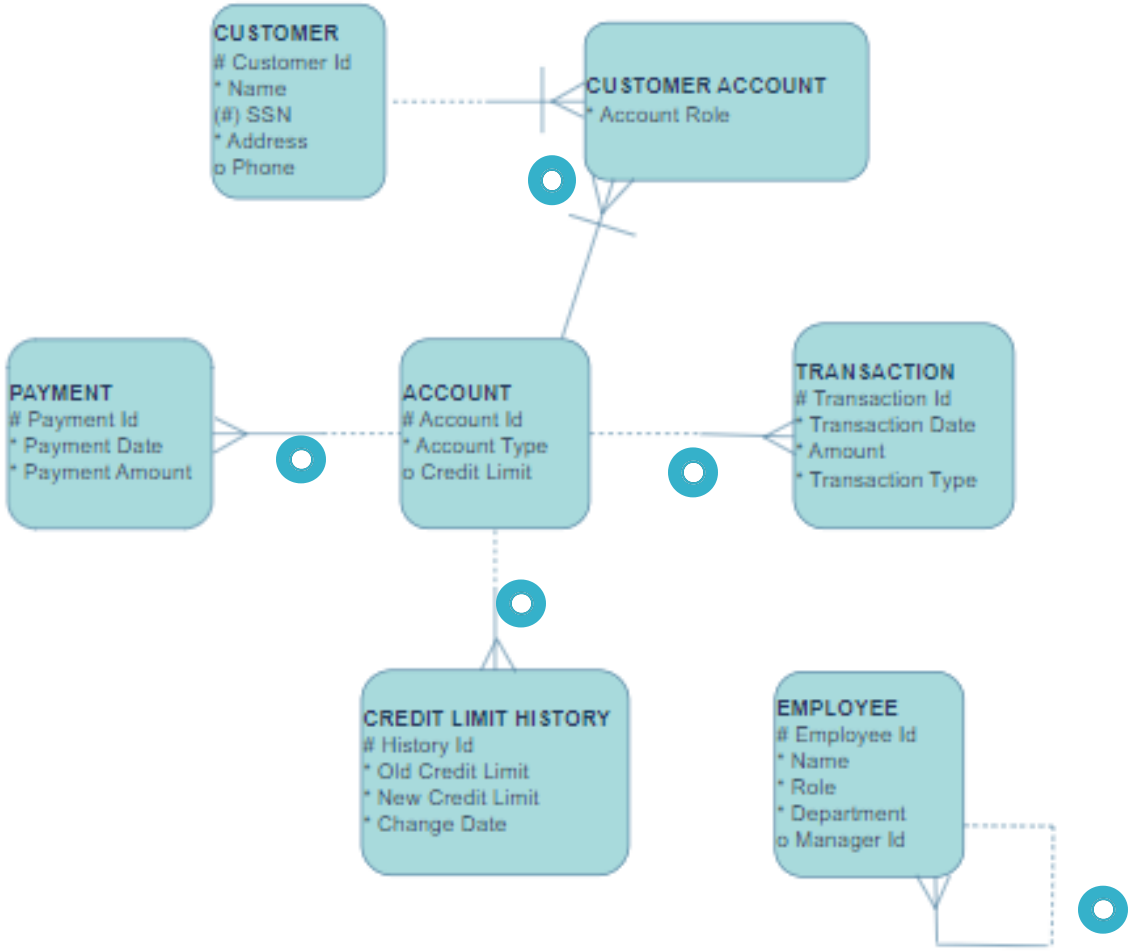




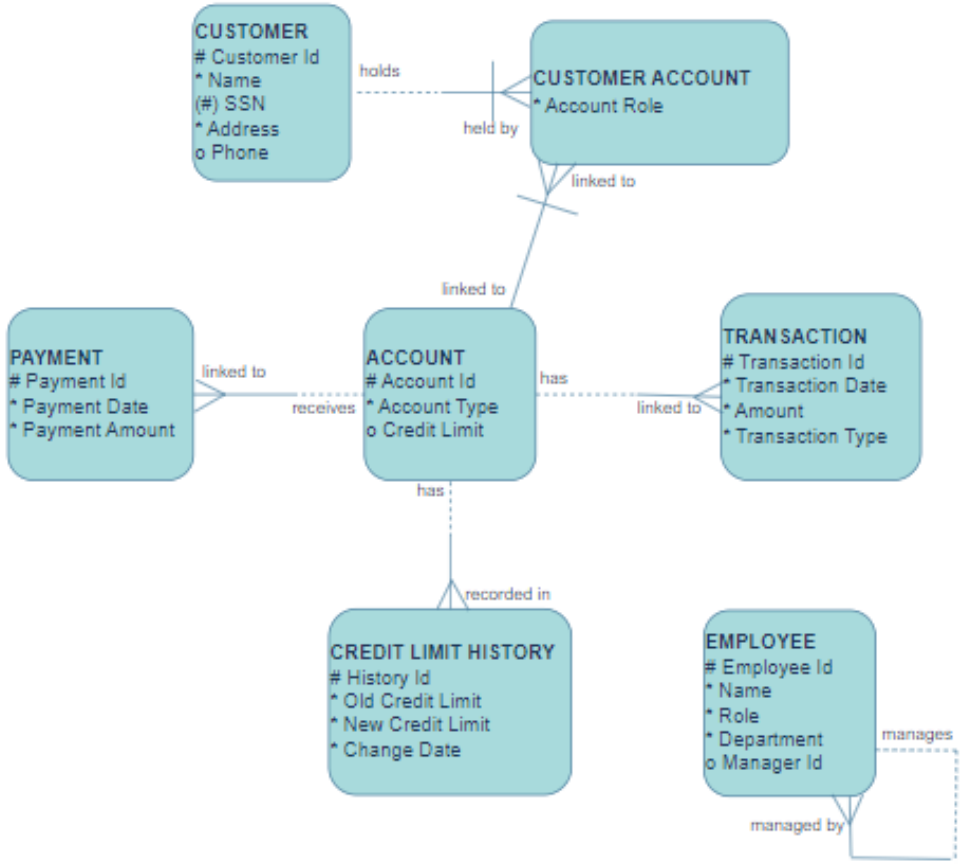


CREDIT LIMIT HISTORY

The goal of Task 8 is to create a table that tracks changes in credit limits for credit accounts over time. The table helps maintain a record of how the credit limit has evolved for an account, providing a historical view of any updates. This ensures that every credit limit adjustment is recorded, allowing SafeBank to track and analyze changes over time, which is important for both operational insights and auditing purposes.


CUSTOMER
Customer Id
* Name
(#) SSN
* Address
o Phone
CUSTOMER ACCOUNT
* Account Role
PAYMENT
Payment Id
* Payment Date
* Payment Amount
ACCOUNT
Account Id
* Account Type
o Credit Limit
TRANSACTION
Transaction Id
* Transaction Date
* Amount
* Transaction Type
CREDIT LIMIT HISTORY
History Id
* Old Credit Limit
* New Credit Limit
* Change Date
EMPLOYEE
Employee Id
* Name
* Role
* Department
o Manager Id

	Customer	Customer Account	Account	Payment	Transaction	Credit Limit History	Employee
Customer	-	holds	-	-	-	-	-
Customer Account	held by	-	linked to	-	-	-	-
Account	-	linked to	-	receives	has	has	-
Payment	-	-	linked to	-	-	-	-
Transaction	-	-	linked to	-	-	-	-
Credit Limit History	-	-	recorded in	-	-	-	-
Employee	-	-	-	-	-	-	manages / managed by



KEY

Attributes

Primary Key: hashtag

Mandatory: asterisk

Optional: lowercase o

Relationships

One and only one: no crow's foot

One or more: crow's foot

Mandatory: solid line

Optional: dashed line

NORMALIZATION

- The database was organized in three steps to reduce data redundancy. First, in the 1NF, we made sure each table had only single values for each attribute, avoiding any repeating groups (e.g., storing multiple phone numbers as separate rows instead of in one field). Then, in the 2NF, we checked that all non-primary key attributes depended on the whole primary key, removing any partial dependencies (e.g., ensuring an attribute like "AccountRole" depends on both CustomerID and AccountID in a joint account table). Finally, in the 3NF, we ensured that all non-primary key attributes relied only on the primary key, eliminating any transitive dependencies (e.g., making "TransactionDate" dependent directly on TransactionID, not indirectly through AccountID). This whole process helped create a cleaner and more efficient database.

DATA INTEGRITY RULES

- Credit Limit Rule: Ensure that the CreditLimit is greater than or equal to 0.
- Transaction Amount Rule: Ensure that the TransactionAmount is greater than 0.
- Payment Amount Rule: Ensure that PaymentAmount is greater than 0.
- SSN Rule: Ensure that the SSN is unique for each customer.

THANKS!